

Question 1: What is a random variable in probability theory?

A random variable is a variable whose value depends on the outcome of a random experiment. It assigns a numerical value to each possible outcome in the sample space. For example, when tossing a die, the random variable can take values from 1 to 6.

Question 2: What are the types of random variables?

There are two main types of random variables: Discrete random variables, which take countable values (e.g., number of students in a class), and Continuous random variables, which take values over a continuous range (e.g., height or weight of a person).

Question 3: Explain the difference between discrete and continuous distributions.

Discrete distributions deal with discrete random variables and have countable outcomes, such as the number of heads in coin tosses. Continuous distributions deal with continuous random variables and have infinite possible values within a range, such as temperature.

Question 4: What is a binomial distribution, and how is it used in probability?

A binomial distribution represents the probability of obtaining a fixed number of successes in a fixed number of independent trials, where each trial has only two outcomes (success or failure). It is commonly used in experiments like coin tosses.

Question 5: What is the standard normal distribution, and why is it important?

The standard normal distribution is a normal distribution with a mean of 0 and a standard deviation of 1. It is important because it allows us to compare different datasets using Z-scores and is widely used in hypothesis testing.

Question 6: What is the Central Limit Theorem (CLT), and why is it critical in statistics?

The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases, regardless of the population distribution. It is critical because it allows statistical inference using normal distribution assumptions.

Question 7: What is the significance of confidence intervals in statistical analysis?

Confidence intervals provide a range of values within which a population parameter is likely to lie with a certain level of confidence (e.g., 95%). They help measure the reliability of estimates.

Question 8: What is the concept of expected value in a probability distribution?

The expected value is the weighted average of all possible values of a random variable, where the weights are their probabilities. It represents the long-term average outcome of a random experiment.

Question 9: Python Code and Output

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.normal(50, 5, 1000)
print("Mean:", np.mean(data))
print("Standard Deviation:", np.std(data))

plt.hist(data, bins=30)
plt.show()
```

Question 10: Explanation and Python Code

Using the Central Limit Theorem, we assume the sampling distribution of the mean is normal. We compute the mean and standard error, then use $Z = 1.96$ for a 95% confidence interval.

```
import numpy as np
import scipy.stats as stats

sales = np.array([220, 245, 210, 265, 230, 250, 260, 275, 240, 255,
                  235, 260, 245, 250, 225, 270, 265, 255, 250, 260])

mean = np.mean(sales)
std_err = stats.sem(sales)
ci = stats.norm.interval(0.95, mean, std_err)

print("Mean Sales:", mean)
print("95% Confidence Interval:", ci)
```